Ch1	5.1	Limits & derivative concept
	5.3	Power rule (integers)
	5.6	Derivatives of x^n (n∈ℚ), sin, cos, e^x, ln x; rules
Ch2	5.9	Kinematics
Ch3	3.4	Circle measure
	3.5	Trig via unit circle; tan=sin/cos; sine rule ambiguous case
	3.8	Solve trig equations
Ch4	5.4	Tangents & normals
	5.2	Increasing/decreasing
	5.7	Second derivative & graphs
	5.8	Extrema & inflection
Ch5	3.6	Pythagorean and double-angle identities
	3.2	Right-triangle trig; sine/cosine rules; area
Ch6	4.1	Data & sampling
	4.2	Presenting data
	4.3	Central tendency & dispersion
Ch7	3.7	Circular functions & transformations
Ch8	4.5	Sample space & simple probability
	4.6	Combined/conditional/independent
	4.11	Conditional probability & independence (revisit)
Ch9	5.5	Intro to integration (power forms)
	5.10	Indefinite integrals & reverse chain rule

Ch10	4.7	Discrete random variables
	4.8	Binomial distribution
Ch11	5.11	Definite integrals & areas
Ch12	4.9	Normal distribution
Ch13	4.10	Regression x on y
	4.12	Standardization (z-scores)

	Start	End
Idea of limit; derivative as gradient and rate of change.		
If f(x)=ax^n then f′(x)=anx^{n−1}; extend to integer exponents.		
Sum/multiple; chain, product, quotient rules.		
s–v–a relations; displacement and total distance via integrals.		
Radians; arc length; sector area.		
Define sin, cos; tan; ambiguous case of sine rule.		
Solve within intervals; quadratics in sinx/cosx/tanx.		
Equations at a point using derivatives.		
Interpret f′>0, f′=0, f′<0.		
Relate f, f′, f′′ to shape (concavity, inflection).		
Local max/min tests; optimization; inflection with zero/non-zero gradients.		
cos^2θ+sin^2θ=1; double-angle identities.		
Use trig ratios; sine/cosine rules; triangle area.		
Population, sample, discrete/continuous; reliability/bias; outliers; techniques.		
Frequency tables, histograms; cumulative frequency; box plots; medians, quartiles, percentiles, range, IQR.		
Mean/median/mode; grouped-data estimates; modal class; SD/variance; effects of transformations; quartiles (discrete).		
sin x, cos x, tan x; amplitude/period/phase/vertical shift; a sin(b(x+c))+d.		
Trial/outcome; equally likely; events; complements; expected number of occurrences.		
Venn, tree, sample-space tables; mutually exclusive; conditional probability; independence.		
Formalize conditional/independent events beyond 4.6.		
Antiderivatives for sums of ax^n (n≠−1); boundary conditions; definite integrals with technology; area with f(x)>0.		
Standard antiderivatives (x^n, sin, cos, e^x, ln x); composites; reverse chain rule.		

Probability distributions; expected value.		
$X \sim B(n,p)$; mean np and variance $np(1-p)$.		
Analytical evaluation; areas with sign; areas between curves.		
Properties; 68–95–99.7 rule; normal probability and inverse normal using technology (SL assumptions).		
Equation and prediction; caution when predicting y from x using x-on-y line.		
Standardized variables; inverse normal with unknown μ,σ contexts.		

Link to 5.12 HL (first principles), 2.4 (inverses/monotonicity).
Link to 5.6 (extended derivatives).
Connect to 5.7–5.9 (behaviour, kinematics).
Links: physics; 4.x (data fitting of motion).
Connect to 3.7 (period), 5.17 HL (volumes).
Link to 3.7 (graphs), 5.6 (derivatives).
Links: 2.10 (solving), 5.8 (stationary points with trig).
Connect to 2.1 (lines), 3.7 (trig slopes).
Connections: 5.8 (optimization), 2.11 (transformations).
Links: 5.8 (max/min tests).
Connect to 2.6–2.11 (features/transformations).
Links: 3.10 HL (compound), 5.15 HL (derivatives of trig family).
Connect to 3.7 (trig graphs), 5.6 (derivatives of sin/cos).
Links: 4.2 (presentation), 4.4 (correlation).
Connect to 4.3 (measures), IA exploratory data analysis.
Links: 4.2 (graphs), 4.10 (regression).
Connect to 2.11 (transformations), 5.6 (chain rule).
Links: 4.8 (binomial), 4.11 (conditional independence).
Link to 4.13 HL (Bayes' theorem).
Link to 4.13 HL (Bayes).
Links to 5.11 (analytical definite integrals).
Links: 5.11 (definite integrals).

5.15 More derivatives (HL)

3.9 Reciprocal trig, identities; inverses (HL)

3.10 Compound-angle identities (HL)

3.11 Trig relationships & symmetry (HL)

5.12 Continuity/differentiability; limits; higher derivative

5.13 Limits & L'Hôpital (HL)

5.14 Implicit differentiation; related rates; optimization

1.16 Linear systems

3.12 Vectors: concepts & proofs (HL)

3.13 Scalar product (HL)

4.13 Bayes' theorem (HL)

4.14 Variance/continuous RVs (HL)

Connect to 4.8 (binomial), 4.14 HL (variance formulas).
Connections: link to 4.5 expected counts; use technology for probabilities; TOK/IM notes.
Connections to 5.5 (technology), 4.14 HL (continuous RV mean).
Links to sciences; modelling cautions.
Connect to 4.4 (y on x line), IA modelling.
Link to 4.9 (normal), 4.10 (regression residuals).

3.14	Line equations (HL)
3.15	Line relationships (HL)
3.16	Vector product (HL)
3.17	Planes (HL)
3.18	Intersections & angles (HL)
5.18	Differential equations (HL)
1.11	Partial fractions
5.16	Integration techniques (HL)
5.17	Areas & volumes (HL)
5.19	Maclaurin series (HL)

3.1	3D distance & midpoint; solids; angles
3.3	Applications & modelling

2.15 Solving $g(x) \geq f(x)$ (HL)

4.4 Bivariate linear correlation

Derivatives of tan, sec, csc, cot, a^x, $\log_a x$, arcsin, arccos, arctan; related integrals; linear composites; partial fractions to prep integration	HL only	Connect to 3.9–3.11 (trig identities), 1.11 HL (partial fractions)
secθ, cscθ, cotθ; identities; arcsin, arccos, arctan (domains/ranges/graphs).	HL only	Connect to 3.10 HL (compound), 5.15 HL (derivatives/integrals).
Formulas for sin(A±B), cos(A±B), tan(2x).	HL only	Links: 3.7 (modelling), 5.15 HL (differentiation/integration).
Graph symmetries and functional relations.	HL only	Connect to 2.14 HL (odd/even), 5.6 (derivatives).
Informal continuity/differentiability; convergence/divergence; higher derivatives.	HL only	Connect to 5.13 (L'Hôpital).
Limit evaluation including repeated L'Hôpital.	HL only	Connect to 5.12 (continuity) and 2.9 (exp/log limits).
Differentiate implicit relations; rates via chain rule; advanced optimization.	HL only	Link to 3.x (geometry), 4.x (modelling).
Solve up to 3×3 systems: unique, infinite, none.	HL only	Links: 2.1–2.4 (lines/intersections), 5.17 HL (areas/volumes setup)
Position/displacement vectors; i,j,k; components/magnitude; geometric proofs.	HL only	Links: 3.13–3.18 HL; physics kinematics.
Dot product and angle between vectors.	HL only	Connect to 3.14–3.18 HL (lines/planes).
Bayes for up to three events; tree/partition forms.	HL only	Connections to 4.6; decision theory.
Variance of discrete/continuous RVs; PDFs; mean/variance/SD; linear transformations.	HL only	Connect to calculus integrals (5.11), z-scores (4.12).

Vector equation r=a+λb; angle between lines; kinematics.	HL only	Link to 3.18 HL (intersections).
Coincident/parallel/intersecting/skew; points of intersection.	HL only	Connect to 3.14 HL (lines), 3.17 HL (planes).
Cross product; properties; |v×w| geometry.	HL only	Links: 3.17 HL (plane normals), 3.18 HL (angles).
r=a+λb+μc; r·n=a·n; Cartesian ax+by+cz=d.	HL only	Connect to 3.18 HL (line/plane intersections).
Line–plane, plane–plane intersections; angles between line–plane and planes.	HL only	Links: 3.14 HL (lines), 5.17 HL (applications).
First-order DEs; Euler's method; separable; homogeneous via y=vx; linear y'+P(x)y=Q(x) using integrating factor.	HL only	Links to 4.14 HL (continuous RV PDFs), physics modelling.
Decompose rational functions.	HL only	Links: 2.13 HL (rational functions), 5.16 HL (integration with partial fractions).
Substitution; integration by parts; repeated parts.	HL only	Links: 1.11 HL (partial fractions), 2.13 HL (rational).
Areas w.r.t. y-axis; volumes of revolution about x or y.	HL only	Connect to 3.1 (solids) and 3.17 HL (planes).
Series for e^x, sin x, cos x, $\ln(1+x)$, $(1+x)^p$; obtain others via substitution/products/integration/differentiation.	HL only	Link to 1.9 (binomial), 1.13 (Euler form).
Distance/midpoint in 3D; area/volume formulas; angle line–line/line–plane.	SL/HL	Links: 3.12–3.18 HL (vectors/planes), 5.17 HL (volumes of revolution).
Angles of elevation/depression; Pythagoras; diagramming from text.	SL/HL	Links: Physics contexts; 4.x data problems.

Inequalities between functions analytically/graphically.	HL only	Link to 2.10 (solve), 5.8 (sign charts for extrema).
PMCC r; scatter plots; line of best fit (by eye, through mean point); regression y on x; interpret a, b	SL/HL	Connect to 4.10 (x on y), IA regression.